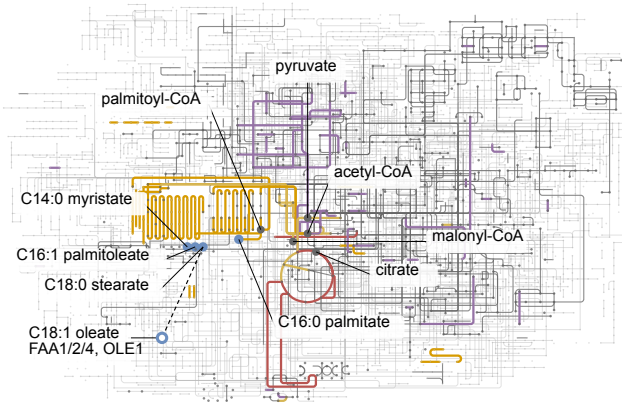




## the route from central carbon to the measured lipids, on KEGG's yeast map

- glycolysis and gluconeogenesis
  - pyruvate metabolism
  - citrate cycle
  - fatty acid biosynthesis, elongation and desaturation
  - fatty acid degradation
  - measured species, a node of the map
  - measured species attached from the yeast GEM
  - named intermediate
  - reaction or compound in Yeast9
  - drawn by KEGG, not in Yeast9
  - - - the GEM's shortest path from a drawn compound; its genes under the name
- The map draws 814 yeast genes, 745 in Yeast9; 889 of its 3878 reaction lines are in Yeast9. Oleate from stearate is attached along the GEM's shortest path: the map has no node for it. Of the 10 deleted regulators, 0 draw a reaction here; of the 13 pathway genes of b, 13 do.